UNIVERSITY OF BIRMINGHAM

School of Physics and Astronomy

DEGREE OF B.Sc. & M.Sci. WITH HONOURS

THIRD-YEAR FINAL EXAMINATION

03 00971

LH RADIATION AND RELATIVITY

SUMMER EXAMINATIONS 2025

Time allowed: 1 hour 30 minutes

Answer ***two*** questions. If you answer more than two questions, credit will only be given for the best two answers.

The approximate allocation of marks to each part of a question is shown in brackets **[ ]**.

Calculators may not be used in this examination.

A formula sheet and a table of physical constants and units that may be required will be found at the end of this question paper.

Answer **two** *questions. If you answer more than two questions, credit will only be given for the best two answers.*

1. In this question, a particle moves along the x-axis with 3-velocity v. You may work in $1+1$ spacetime, i.e. all y- and z-coordinates can be ignored.

   (a) In the context of spacetime interval $(\mathrm{d}s)^2$ between two spacetime events, explain the meaning of *timelike-*, *spacelike-* and *null-separated* events. From the invariance of $(\mathrm{d}s)^2$, show that $\mathrm{d}t = \gamma(v)\mathrm{d}\tau$. What kind of separation must these events have? Therefore show that the 4-velocity of a material particle, $u^\mu = \mathrm{d}x^\mu/\mathrm{d}\tau$, is related to its 3-velocity, $v^i = c\beta^i = \mathrm{d}x^i/\mathrm{d}t$, by **[7]**

$$u^\mu \;\equiv\; (u^0, u^i) \;=\; \gamma(v)(c, v^i) \;\equiv\; c\gamma(v)(1, \beta^i).$$

   (b) A particle moves along the x-axis velocity v. Express the (ct, x) coordinates of the particle's 4-velocity in terms of the particle's rapidity, $w = \operatorname{arctanh}\beta$. Find an expression for the proper time derivative $\mathrm{d}w/\mathrm{d}\tau$ in terms of the 3-acceleration, $a = \mathrm{d}v/\mathrm{d}t$. Hence find an expression for the 4-acceleration, $\alpha^\mu = \mathrm{d}u^\mu/\mathrm{d}\tau$ in terms of w and a. **[8]**
   (Hint: Recall that $\mathrm{d}\operatorname{arctanh} Z/\mathrm{d}X = \cosh^2 X$*.)*

   (c) Write down α^μ in the particle's comoving frame, Σ', in which $w = 0$. Use this to show that, for the particle accelerated from rest with a constant 3-acceleration a' (as measured in Σ'), the rapidity after time τ is given by **[5]**

$$w(\tau) \;=\; \frac{a'\tau}{c}.$$

The Sirius star system is separated from the earth by approximately 9 light years in their mutual rest frame. We choose their separation to be along the x-direction. A spacecraft leaves the Earth in the year 3000, travelling to Sirius. For the first half of its outward journey, it has a constant acceleration $a' \approx 10\,\mathrm{m\,s^{-2}}$ in the spacecraft's comoving frame, while for the second half it continues with a constant deceleration with the same magnitude.

   (c) Estimate the duration of the journey according to the spacecraft's clock, and the year that the spacecraft arrives at Sirius. **[10]**
   (For numerical estimates use the following facts: $T_0 = c/a' \approx 1$ *year; and* $\operatorname{arccosh} 5.5 \approx \operatorname{arcsinh} 5.5 \approx 2.4$*.)*

2. (a) Write down expressions for electric field $\boldsymbol{E}$ and magnetic field $\boldsymbol{B}$ in terms of the electromagnetic scalar potential V and vector potential $\boldsymbol{A}$, and show that these expressions are consistent with the Faraday tensor identity $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$. Write down the physical dimensions (products of `[Length]`, `[Time]`, `[Mass]`, `[Charge]`) of $\boldsymbol{E}$, $\boldsymbol{B}$, $\boldsymbol{A}$ and V. **[8]**

Now assume that we can define a *dual 4-potential* $C^\mu = (U, \boldsymbol{C})$ which is related to the dual Faraday tensor by

$$\partial_\mu C_\nu - \partial_\nu C_\mu \;=\; G_{\mu\nu}.$$

(b) What are the dimensions of the components of C^μ? Find expressions for $\boldsymbol{E}$ and $\boldsymbol{B}$ in terms of U and $\boldsymbol{C}$. By considering Maxwell's equations, describe under what physical conditions C^μ exists. **[9]**

(c) Describe and explain the significance of the *Lorenz gauge* for the 4-potential A^μ. Assuming the physical conditions are satisfied for it to exist, describe the analogue of the Lorenz gauge for the dual 4-potential C^μ. **[7]**

(d) Hence or otherwise write down the dual 4-potential for a circular polarised electromagnetic plane wave. How does this differ from the standard 4-potential for a circularly polarised plane wave? **[6]**

3. The Liénard-Wiechert potentials due to a particle of charge q moving along a trajectory $\boldsymbol{r}''(t)$ are given by

$$V(\boldsymbol{r}, t) \;=\; \frac{1}{4\pi\varepsilon_0}\frac{q}{(R - \boldsymbol{R}\cdot\boldsymbol{\beta})}, \qquad \boldsymbol{A}(\boldsymbol{r}, t) \;=\; \frac{1}{4\pi\varepsilon_0}\left[\left[\frac{q\boldsymbol{\beta}}{c(R - \boldsymbol{R}\cdot\boldsymbol{\beta})}\right]\right],$$

where $\boldsymbol{R} = \boldsymbol{r}'' - [[\boldsymbol{r}'']]$, 3-velocity is $c\boldsymbol{\beta}$, and $[[\bullet]]$ denotes quantities evaluated at retared time.

(a) Sketch a diagram of a particle worldline and suitable light cones to demonstrate the definitions of events x'' and $[[x'']]$. Explain what is meant by *retarded time* t'' and derive a formula for t'' in terms of $\boldsymbol{r}''$, $[[\boldsymbol{r}'']]$ and t. **[8]**

(b) By considering the four-potential, A^μ, in the comoving frame of the charge, derive the above formulas for the Liénard-Wiechert potentials. **[8]**

The electric and magnetic fields due to the Liénard-Wiechert potentials are

$$\boldsymbol{E} \;=\; \frac{1}{4\pi\varepsilon_0}\left[\left[\frac{q(\boldsymbol{e}_R - \boldsymbol{\beta})}{R^2\gamma^2(1 - \boldsymbol{e}_R\cdot\boldsymbol{\beta})^3}\right]\right] + \frac{1}{4\pi\varepsilon_0}\left[\left[\frac{q\boldsymbol{e}_R \times \{(\boldsymbol{e}_R - \boldsymbol{\beta})\times\boldsymbol{a}\}}{Rc^2(1 - \boldsymbol{e}_R\cdot\boldsymbol{\beta})^3}\right]\right],$$

$$\boldsymbol{B} \;=\; c^{-1}[[\boldsymbol{e}_R]] \times \boldsymbol{E},$$

where $\boldsymbol{e}_R \;=\; \boldsymbol{R}/R$.

(c) Show that the total power radiated by the charge in the non-relativistic limit is

$$\frac{\mathrm{d}\mathcal{E}}{\mathrm{d}t} = \frac{q^2 a^2}{6\pi\varepsilon_0 c^3},$$

where $\boldsymbol{a}$ is the acceleration of the charge. **[8]**

(Hint: recall that $\int_0^\pi \sin^3 X \mathrm{d}X = 4/3$*)*

(d) A system of point particles, indexed by i, with charge q_i and 3-positions $\boldsymbol{r}_i$ have net dipole moment is $\boldsymbol{d} \equiv \sum_i q_i \boldsymbol{r}$. Show that the radiation field at large distances, in the non-relativistic limit due to an accelerating dipole moment $\boldsymbol{d}(t)$, is **[6]**

$$\boldsymbol{E} = \frac{1}{4\pi\varepsilon_0 c^2 R}(\ddot{\boldsymbol{d}} \times \boldsymbol{e}_R) \times \boldsymbol{e}_R, \qquad \boldsymbol{B} = \frac{\mu_0}{4\pi\varepsilon_0 c R}(\ddot{\boldsymbol{d}} \times \boldsymbol{e}_R).$$

Formula Sheet

- **Maxwell's equations** in 3-form, for electric field $\boldsymbol{E}$, magnetic field $\boldsymbol{B}$, electric current density $\boldsymbol{J}$ and charge density ρ, in empty space, with electric constant ε_0 and magnetic constant μ_0, where $c = 1/\sqrt{\varepsilon_0\mu_0}$,

$$\begin{aligned} \nabla\cdot\boldsymbol{B} &= 0, & \nabla\times\boldsymbol{E}+\dot{\boldsymbol{B}} &= 0, \\ \varepsilon_0\nabla\cdot\boldsymbol{E} &= \rho, & \mu_0^{-1}\nabla\times\boldsymbol{B}-\varepsilon_0\dot{\boldsymbol{E}} &= \boldsymbol{J}, \end{aligned}$$

- **Lorentz force equations** in in 3-form, for particle charge q, speed $\boldsymbol{v}$, momentum $\boldsymbol{p}$, and energy $\mathcal{E}$,

$$\frac{\mathrm{d}\boldsymbol{p}}{\mathrm{d}t} = q(\boldsymbol{E}+\boldsymbol{v}\times\boldsymbol{B}), \qquad \frac{\mathrm{d}\mathcal{E}}{\mathrm{d}t} = q\boldsymbol{E}\cdot\boldsymbol{v}.$$

- **Lorentz transformed EM fields** after a Lorentz boost by velocity $\boldsymbol{v}$,

$$\begin{aligned} \overline{\boldsymbol{E}} &= \gamma(\boldsymbol{E}+\boldsymbol{v}\times\boldsymbol{B}) - \frac{\gamma^2}{c^2(\gamma+1)}\boldsymbol{v}(\boldsymbol{v}\cdot\boldsymbol{E}), \\ \overline{\boldsymbol{B}} &= \gamma(\boldsymbol{B}-c^{-2}\boldsymbol{v}\times\boldsymbol{E}) - \frac{\gamma^2}{c^2(\gamma+1)}\boldsymbol{v}(\boldsymbol{v}\cdot\boldsymbol{B}) \end{aligned}$$

- **Electromagnetic potentials** with V the scalar potential, $\boldsymbol{A}$ the vector potential,

$$\boldsymbol{B} = \nabla\times\boldsymbol{A}, \qquad \boldsymbol{E} = -\dot{\boldsymbol{A}} - \nabla V.$$

- **Minkowski Metric**, given by $\eta_{\mu\nu}$ or $\eta^{\mu\nu}$, where all components zero except

$$\eta_{00} = \eta^{00} = +1, \qquad \eta_{jj} = \eta^{jj} = -1, \qquad \text{for } j = 1,2,3.$$

- **Spacetime position** x^μ transforms as contravariant 4-vector,

$$x^\mu = (x^0, x^1, x^2, x^3) = (ct, \boldsymbol{r}); \qquad x_\mu = (x_0, x_1, x_2, x_3) = (ct, -\boldsymbol{r}).$$

- **Covariant Derivative**, ∂_μ transforms as covariant 4-vector,

$$\partial_\mu = (\partial_0, \partial_1, \partial_2, \partial_3) = (c^{-1}\partial_t, \boldsymbol{\nabla}), \qquad \partial^\mu = (\partial^0, \partial^1, \partial^2, \partial^3) = (c^{-1}\partial_t, -\boldsymbol{\nabla}).$$

- **Lorentz transformation**, $\Lambda^{\overline{\mu}}_{\nu}$ from Σ to $\overline{\Sigma}$, for a boost by velocity $v = \beta c$ in standard configuration, has nonzero components

$$\Lambda^{\overline{0}}_{0} = \Lambda^{\overline{1}}_{1} = \gamma, \quad \Lambda^{\overline{0}}_{1} = \Lambda^{\overline{1}}_{0} = -\gamma\beta, \quad \Lambda^{\overline{2}}_{2} = \Lambda^{\overline{3}}_{3} = 1,$$

  with the Lorentz factor $\gamma = \gamma(v) = (1-\beta^2)^{-1/2}$.

- **d'Alembertian operator**, a Lorentz scalar operator,

$$\Box^2 \equiv \eta^{\mu\nu}\partial_\mu\partial_\nu = c^{-2}\partial_t^2 - \nabla^2.$$

- **Faraday tensor** in matrix form

$$F_{\mu\nu} = \begin{pmatrix} 0 & E_x/c & E_y/c & E_z/c \\ -E_x/c & 0 & -B_z & B_y \\ -E_y/c & B_z & 0 & -B_x \\ -E_z/c & -B_y & B_x & 0 \end{pmatrix},$$

$$F^{\mu\nu} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix},$$

so $F_{0i} = E_i/c$ and $F^{ij} = -\varepsilon^{ijk} B_k$, for $i, j, k = 1, 2, 3$.
It follows from the 4-potential $A^\mu = (V/c, \boldsymbol{A})$ that

$$F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu.$$

- **dual Faraday tensor** $G_{\mu\nu}$ is given by

$$G_{\mu\nu} = \tfrac{1}{2}\varepsilon_{\mu\nu\alpha\beta} F^{\alpha\beta}, \qquad G^{\mu\nu} = \tfrac{1}{2}\varepsilon^{\mu\nu\alpha\beta} F_{\alpha\beta}.$$

Its components satisfy $G_{0i} = -B_i$, $G^{ij} = -\varepsilon^{ijk} E_k/c$ for $i, j, k = 1, 2, 3$.

- **Levi-Civita symbol** in 3D satisfies

$$\varepsilon_{ijk} = \varepsilon^{ijk} = \begin{cases} +1 & \text{if } \; ijk \text{ is an even permutation} \\ -1 & \text{if } \; ijk \text{ is an odd permutation.} \end{cases}$$

Here, even permutations consist of $123, 231, 312$, and odd permutations are $132, 213, 321$. It also satifies $\varepsilon_{ijk}\varepsilon^{imn} = \delta^m_j \delta^n_k - \delta^m_k \delta^n_j$, and

$$\boldsymbol{a} \cdot (\boldsymbol{b} \times \boldsymbol{c}) = \varepsilon^{ijk} a_i b_j c_k.$$

- **Levi-Civita symbol** in 4D satisfies

$$\varepsilon_{\mu\nu\rho\sigma} = -\varepsilon^{\mu\nu\rho\sigma} = \begin{cases} +1 & \text{if } \; \mu\nu\rho\sigma \text{ is an even permutation of } 0123 \\ -1 & \text{if } \; \mu\nu\rho\sigma \text{ is an odd permutation of } 0123 \\ 0 & \text{if } \; \text{anything else.} \end{cases}$$

The signs of the permutations of $0, 1, 2, 3$ are given by

$+$	0123,	0231,	0312,	1032,	1203,	1320,	2013,	2130,	2301,	3021,	3102,	3210,
$-$	0132,	0213,	0321,	1023,	1230,	1302,	2031,	2103,	2310,	3012,	3120,	3201.

Physical Constants and Units

Acceleration due to gravity	g	$9.81\,\mathrm{m\,s^{-2}}$
Gravitational constant	G	$6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$
Ice point	T_{ice}	$273.15\,\mathrm{K}$
Avogadro constant	N_A	$6.022 \times 10^{23}\,\mathrm{mol^{-1}}$
		[*N.B.* 1 mole $\equiv 1$ *gram-molecule*]
Gas constant	R	$8.314\,\mathrm{J\,K^{-1}\,mol^{-1}}$
Boltzmann constant	k, k_B	$1.381 \times 10^{-23}\,\mathrm{J\,K^{-1}} \equiv 8.62 \times 10^{-5}\,\mathrm{eV\,K^{-1}}$
Stefan constant	σ	$5.670 \times 10^{-8}\,\mathrm{W\,m^{-2}\,K^{-4}}$
Rydberg constant	R_∞	$1.097 \times 10^{7}\,\mathrm{m^{-1}}$
	$R_\infty hc$	$13.606\,\mathrm{eV}$
Planck constant	h	$6.626 \times 10^{-34}\,\mathrm{J\,s} \equiv 4.136 \times 10^{-15}\,\mathrm{eV\,s}$
$h/2\pi$	$\hbar$	$1.055 \times 10^{-34}\,\mathrm{J\,s} \equiv 6.582 \times 10^{-16}\,\mathrm{eV\,s}$
Speed of light *in vacuo*	c	$2.998 \times 10^{8}\,\mathrm{m\,s^{-1}}$
	$\hbar c$	$197.3\,\mathrm{MeV\,fm}$
Charge of proton	e	$1.602 \times 10^{-19}\,\mathrm{C}$
Mass of electron	m_e	$9.109 \times 10^{-31}\,\mathrm{kg}$
Rest energy of electron		$0.511\,\mathrm{MeV}$
Mass of proton	m_p	$1.673 \times 10^{-27}\,\mathrm{kg}$
Rest energy of proton		$938.3\,\mathrm{MeV}$
One atomic mass unit	u	$1.66 \times 10^{-27}\,\mathrm{kg}$
Atomic mass unit energy equivalent		$931.5\,\mathrm{MeV}$
Electric constant	ϵ_0	$8.854 \times 10^{-12}\,\mathrm{F\,m^{-1}}$
Magnetic constant	μ_0	$4\pi \times 10^{-7}\,\mathrm{H\,m^{-1}}$
Bohr magneton	μ_B	$9.274 \times 10^{-24}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Nuclear magneton	μ_N	$5.051 \times 10^{-27}\,\mathrm{A\,m^2}$ ($\mathrm{J\,T^{-1}}$)
Fine-structure constant	$\alpha = e^2/4\pi\epsilon_0\hbar c$	7.297×10^{-3}= 1/137.0
Compton wavelength of electron	$\lambda_c = h/m_e c$	$2.426 \times 10^{-12}\,\mathrm{m}$
Bohr radius	a_0	$5.2918 \times 10^{-11}\,\mathrm{m}$
angstrom	Å	$10^{-10}\,\mathrm{m}$
barn	b	$10^{-28}\,\mathrm{m^2}$
torr (mm Hg at $0\,^\circ$C)	torr	$133.32\,\mathrm{Pa}$ ($\mathrm{N\,m^{-2}}$)

Do not complete the attendance slip, fill in the front of the answer book or turn over the question paper until you are told to do so.

Important Reminders

- Coats/outwear should be placed in the designated area.
- Unauthorised materials (e.g. notes or Tippex) must be placed in the designated area.
- Check that you do not have any unauthorised materials with you (e.g. in your pockets, pencil case).
- Mobile phones and smart watches **must** be switched off and placed in the designated area or under your desk. They must not be left on your person or in your pockets.
- You are not permitted to use a mobile phone as a clock. If you have difficulty seeing a clock, please alert an Invigilator.
- You are not permitted to have writing on your hand, arm or other body part.
- Check that you do not have writing on your hand, arm or other body part – if you do, you must inform an Invigilator immediately
- Alert an Invigilator immediately if you find any unauthorised item upon you during the examination.

Any students found with non-permitted items upon their person during the examination, or who fail to comply with Examination rules may be subject to Student Conduct procedures.